

Object Detection Neural Network

Predicate Prediction Neural Network

loss

X, Y

VR Plausibility Rule Engine

for each rule, query the KG for whether objects X and Y belong to rule-specific classes (often high-level parent classes), as inferred by OWL reasoning, given their asserted base classes and OWL ontology VRD-World



knowledge graph hosting NeSy4VRD OWL ontology VRD-World

ordered pair of objects X, Y

assert X, Y base classes in KG: vrd:X rdf:type vrd:ClassX . vrd:Y rdf:type vrd:ClassY .